GDSR: Global-Detail Integration Through Dual-Branch Network With Wavelet Losses for Remote Sensing Image Super-Resolution

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***Abstract*—In recent years, deep neural networks, including convolutional neural networks (CNNs), Transformers, and State-Space Models, have achieved significant progress in remote sensing image (RSI) super-resolution (SR). However, existing SR methods typically overlook the complementary relationship between global and local dependencies. These methods either focus on capturing local information or prioritize global information, which results in models that are unable to effectively capture both global and local features simultaneously. Moreover, their computational cost becomes prohibitive when applied to large-scale RSIs. To address these challenges, we introduce the novel application of receptance weighted key value (RWKV) to RSI-SR, which captures long-range dependencies with linear complexity. To simultaneously model global and local features, we propose the global-detail dual-branch structure, Global-Detail SR (GDSR), which performs SR by paralleling RWKV and convolutional operations to handle large-scale RSIs. Furthermore, we introduce the global-detail reconstruction module (GDRM) as an intermediary between the two branches to bridge their complementary roles. In addition, we propose the dual-group multiscale wavelet loss, a wavelet-domain constraint mechanism via dual-group subband strategy and cross-resolution frequency alignment for enhanced reconstruction fidelity in RSI-SR. Extensive experiments under two degradation methods on several benchmarks, including AID, UCMerced, and RSSRD-QH, demonstrate that GSDR outperforms the state-of-the-art Transformer-based method HAT by an average of 0.09 dB in the peak signal-to-noise ratio (PSNR), while using only 63% of its parameters and 51% of its FLOPs, achieving an inference speed 3.2 times faster.**

***Index Terms*—Dual-branch, global-detail reconstruction, remote sensing image (RSI), super-resolution (SR).**

Received 24 March 2025; revised 3 June 2025 and 15 August 2025; accepted 28 September 2025. Date of publication 6 October 2025; date of current version 16 October 2025. This work was supported in part by the Natural Science Foundation of Qinghai Province under Grant 2023-ZJ-906M; and in part by the National Natural Science Foundation of China under Grant U23A20159, Grant 62162053, and Grant 42265010. *(Corresponding authors: Guojing Zhang; Xiaoying Wang.)*

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Digital Object Identifier 10.1109/TGRS.2025.3617006

I. Introduction

THE rapid development of remote sensing technologies has resulted in an ever-increasing availability of satellite and aerial images, which play a crucial role in a variety of applications, such as environmental monitoring [1], urban planning [2], and disaster management [3]. However, the resolution of these remote sensing images is often constrained by the limitations of imaging sensors, atmospheric interference, and satellite bandwidth. Consequently, SR techniques, which seek to reconstruct high-resolution (HR) images from low-resolution (LR) observations, have garnered significant interest from both academic and industrial sectors [4], [5]. In particular, remote sensing image super-resolution (RSISR) serves as a pivotal technique for enhancing spatial resolution to extract fine-grained details required for precise geospatial interpretation. However, quantitative frequency analysis highlights a fundamental imbalance in spatial frequency composition between LR and HR observations. As illustrated in Fig. 1, LR images predominantly contain low-frequency components, while critical high-frequency (HF) features representing geospatial patterns are severely attenuated. This asymmetry is exacerbated by conventional degradation processes that disproportionately suppress HF spatial information, creating reconstruction bottlenecks that standard SR methods struggle to resolve effectively.

Traditional SR methods have been dominated by convolutional neural networks (CNNs) [6], [7], [8], [9] due to their strong local feature extraction and computational efficiency. However, as depicted in Fig. 2(a), the limited effective receptive field (ERF) [10] of CNNs inherently restricts their ability to model long-range dependencies. This becomes particularly challenging in large-scale remote sensing images (RSIs), where diverse spatial patterns and textures overwhelm the limited modeling capacity of CNN-based methods [11]. While Transformer-based models [11], [12], [13] alleviate the ERF limitation through global self-attention mechanisms, their quadratic computational complexity necessitates window-based self-attention that paradoxically shrinks effective receptive field coverage [as depicted in Fig. 2(b)], and their sequence transformation processes risk losing structural information critical for spatial consistency in RSIs [14]. Recent advances in linear architectures address the computational constraints of Transformers through distinct

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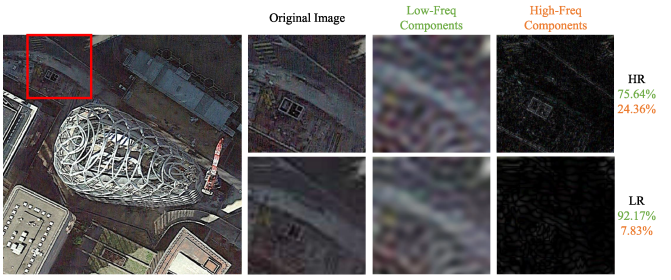

Fig. 1. Example of spatial frequency decomposition between HR and LR RSIs.

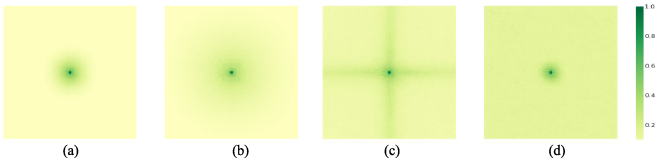

Fig. 2. ERF [10] visualization for different models with different architectures. A more extensively distributed dark area indicates a larger ERF. (a) CNN-based method. (b) Transformer-based method. (c) SSM-based method. (d) RWKV-based method.

approaches. Mamba-based models [15], [16] leverage state-space sequences (SSMs) [17] to achieve linear-complexity long-range dependency modeling, demonstrating performance comparable to Transformers. However, their inherent sequence modeling in SSMs still poses challenges in achieving an optimal ERF for 2-D images [as depicted in Fig. 2(c)], limiting spatial pattern capture in RSIs. Conversely, receptance weighted key value (RWKV)-based models combine linear attention mechanisms with localized dependency modeling, outperforming both Transformers and Mamba variants in vision tasks [18], [19], [20] and offering efficient long-range modeling with expanded ERF [as depicted in Fig. 2(d)]. Despite pioneering efforts to adopt RWKV for vision tasks, the adaptation of RWKV to the field of RSISR remains unexplored. Crucially, existing approaches spanning CNNs, Transformers, Mamba, and RWKV variants universally neglect the complementary synergy between global attention mechanisms (low-pass filtering) and convolutional operations (high-pass filtering) [21], failing to synergize structural context preservation with HF detail recovery essential for precise large-scale RSI reconstruction. This gap underscores the need for architectures that unify efficient global dependency modeling and localized texture refinement tailored to RSISR.

In this article, we propose a novel approach that integrates global representations and detailed extraction with wavelet losses to enhance the performance of RSISR. Our main contributions can be summarized as follows.

1) Global-detail SR (GDSR) is proposed to synergistically unify the RWKV and CNN in a dual-branch framework, where global contextual dependencies and localized spatial features are collaboratively refined for RSISR.
2) A global-detail reconstruction module (GDRM) is proposed to integrate information from the dual branches efficiently. By merging complementary features, this module enhances the synergy between the branches, thereby significantly improving the overall performance of RSISR tasks.
3) Dual-group multiscale wavelet loss is proposed to unify low-frequency structural subbands and aggregated HF detail subbands through a dual-coefficient grouping strategy, enforcing cross-resolution frequency consistency via hierarchical wavelet decomposition across resolutions in a multiscale framework, leveraging wavelet-domain frequency priors to augment spatial-domain optimization.

We demonstrate the empirical validity of GDSR by comparing it with various competitive CNN-, GAN-, Transformer-, Mamba-, and hybrid-based models on three RSI benchmark datasets using two degradation methods, showing SOTA results. In addition, we conducted extensive ablation experiments (as detailed in Section IV-C) to investigate the approach of global-detail integration and its effect on RSI-SR and to examine the effect of dual-group multiscale wavelet loss. We believe our work will advance efficient large-scale RSI-SR, improving both fidelity and processing speed, while ensuring robustness and practical applicability for real-world remote sensing tasks.

II. Related Work

A. Remote Sensing Image SR

Recent advances in deep learning have greatly improved RSI-SR, with methods typically falling into CNN-, Transformer-, and Mamba-based categories. CNN-based methods [22], [23], [24], [25], inspired by SRCNN [6], have used modules like residual connections and attention mechanisms to enhance performance. However, they are limited by their inability to capture long-range dependencies, which hinders their effectiveness in large-scale RSI data. Transformer-based methods [11], [26], [27] leverage self-attention (SA) to overcome CNN's locality constraints. While standard SA achieves global interaction through pairwise token correlation, its quadratic complexity becomes prohibitive for HR RSIs. Window-based SA [12] reduces computation via local window partitioning, yet critically limits cross-window dependencies required for reconstructing continuous geographic features (e.g., highway systems or river networks) spanning multiple windows. Mamba-based methods emerge as an alternative with linear complexity. MambaIR [15] successfully introduced Mamba [17] into the SR domain, while FreMamba [16] pioneers the exploration of Mamba's potential for RSI-SR, extending it with frequency analysis. However, SSMs' causal modeling framework, constrained by 1-D sequence scanning of 2-D spatial data, fundamentally struggles to capture omnidirectional contextual relationships required for coherent reconstruction of geometrically complex remote sensing features (e.g., roundabouts with radial roads). Emerging hybrid architectures combining multiple paradigms have demonstrated significant progress. Recent works [28], [29], [30], [31] propose innovative solutions integrating CNN–Transformer feature fusion, U-Net structures with adaptive edge recovery, and shift channel attention-enhanced Transformers, effectively addressing limitations of

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single-paradigm approaches while maintaining computational efficiency.

B. RWKV in Computer Vision

Recent studies [18], [19], [32] highlight the superiority of RWKV over Transformers and Mamba in NLP tasks due to its linear computational complexity and parallelizable recurrent architecture. In computer vision, Vision-RWKV [33] adapts RWKV for 2-D spatial processing by introducing a bidirectional WKV attention mechanism to capture global dependencies and a quad-directional token shift operation to model local contextual relationships across horizontal and vertical axes. These adaptations preserve the efficiency of RWKV while enabling effective feature learning in images. Subsequent developments further expand RWKV's applicability. Diffusion-RWKV [34] integrates RWKV into diffusion models for image generation, leveraging bidirectional processing to handle patched image inputs through linear-complexity spatial aggregation, thereby eliminating the need for window-based operations. Restore-RWKV [20] enhances 2-D dependency modeling through a recurrent WKV (Re-WKV) mechanism that integrates bidirectional and recurrent attention across multiple scan directions, alongside an omnidirectional token shift (Omni-Shift) layer to broaden context aggregation.

However, existing implementations remain suboptimal for RSI due to distinct domain characteristics. Restore-RWKV's homogeneous U-Net backbone suits anatomical medical data but falters with RSI's extreme scale variations and heterogeneous geospatial textures. This article introduces a dual-branch framework that decouples global feature extraction using RWKV from local feature extraction using CNN, directly addressing RSI-SR-specific challenges unmet by homogeneous architectures.

C. WTs in Deep Learning

The wavelet transform (WT) [35] has been a fundamental signal-processing tool since the 1980s and has recently been integrated into neural network architectures. The initial wave of deep learning integration saw wavelet-based architectures like Wavelet-SRNet [36], WDST [37], and WDRN [38] employing wavelet coefficient prediction for SR enhancement. Subsequent innovations deepened this synergy, with WTConv [39] introducing trainable wavelet-based layers that dynamically expand convolutional receptive fields, demonstrating cross-task adaptability in computer vision. The integration further evolved through adversarial frameworks, particularly WGSR [40] that embedded wavelet subband losses within GAN architectures. Moreover, [41] extended this strategy by integrating wavelet and RGB losses in Transformer-based models, enhancing HF preservation and achieving superior visual performance on natural images compared to pixel-wise loss training.

Notably, existing wavelet-based SR methods typically process subbands at a single scale, failing to capture the multiscale characteristics intrinsic to RSI textures. This fragmented optimization approach separately constrains low-frequency structural components and HF directional details without explicit cross-scale coordination, leading to inconsistent reconstruction of hierarchical spatial patterns. In this article, we propose a novel dual-group multiscale wavelet loss to enforce cross-resolution frequency consistency through explicit approximation subband grouping and multiscale HF aggregation (horizontal/vertical/diagonal).

III. Methodology

In this section, we first introduce the overall structure of the proposed GDSR and then describe three important modules in GDSR, namely RGEG, RDEG, and GDRM. Finally, we detail our designed loss function.

A. Overview of GDSR

As illustrated in Fig. 3, GDSR begins by extracting the shallow feature $\boldsymbol{F}_S \in \mathbb{R}^{H\times W\times C}$ from the LR input using a convolutional layer. It then derives the deep features $\boldsymbol{F}_{D_g} \in \mathbb{R}^{H\times W\times C}$ and $\boldsymbol{F}_{D_d} \in \mathbb{R}^{H\times W\times C}$ through parallel branches: global extraction branch and details extraction branch, respectively. These features are subsequently fused using the GDRM. A global residual connection is then applied to combine the low-level and deep features, and finally, the pixel-shuffle method [42] is employed for upsampling to generate the SR output. The SR process of GDSR can be mathematically described as

$$\boldsymbol{I}_{\mathrm{SR}} = \mathrm{Up}(\mathrm{GDRM}(\boldsymbol{F}_{D_g}, \boldsymbol{F}_{D_d}) + \boldsymbol{F}_S) \tag{1}$$

where $\mathrm{Up}(\cdot)$ denotes the upsampling function and $\mathrm{GDRM}(\cdot)$ denotes the global-detail reconstruction operation. $\boldsymbol{I}_{\mathrm{SR}} \in \mathbb{R}^{(rH)\times(rW)\times 3}$ is the reconstructed SR image with a scaling factor of r.

B. Residual Details Extraction Group

Deep convolutional networks have achieved remarkable advancements in SR tasks, with residual structures proving effective in addressing network degradation issues [9]. To harness these advantages, we propose RDEGs, which are structured as sequences of residual units, each referred to as a residual convolution block (RCB). RCBs are specifically designed to enhance the recovery of fine details from LR inputs while maintaining stable and efficient model training.

A single RCB can be represented as

$$\boldsymbol{F}_{l+1} = \alpha \boldsymbol{F}_l + W_1(\mathrm{LReLU}(W_0 \boldsymbol{F}_l)) \tag{2}$$

where $\boldsymbol{F}_l$ and $\boldsymbol{F}_{l+1}$ represent the input and output feature maps of the lth RCB, respectively. α denotes a learnable scale factor that controls the amount of information passed through the residual connection. W_0 and W_1 denote the convolution operations in the RCB and LReLU refers to the Leaky ReLU activation function.

We concatenate N RCBs to form an RDEG module. The RDEG can be mathematically expressed as

$$\mathrm{RDEG}^k = \alpha \boldsymbol{F}_{D_d}^{N-1} + W_1\left(\mathrm{LReLU}\left(W_0\left(\boldsymbol{F}_{D_d}^{N-1}\right)\right)\right), \quad N \geq 1 \tag{3}$$

where $\boldsymbol{F}_{D_d}^0 = \mathrm{RDEG}^{k-1}$. Here, RDEG^k represents the feature map of the kth RDEG and $\boldsymbol{F}_{D_d}^{N-1}$ denotes the output of the $(N-1)$th RCB within the kth RDEG.

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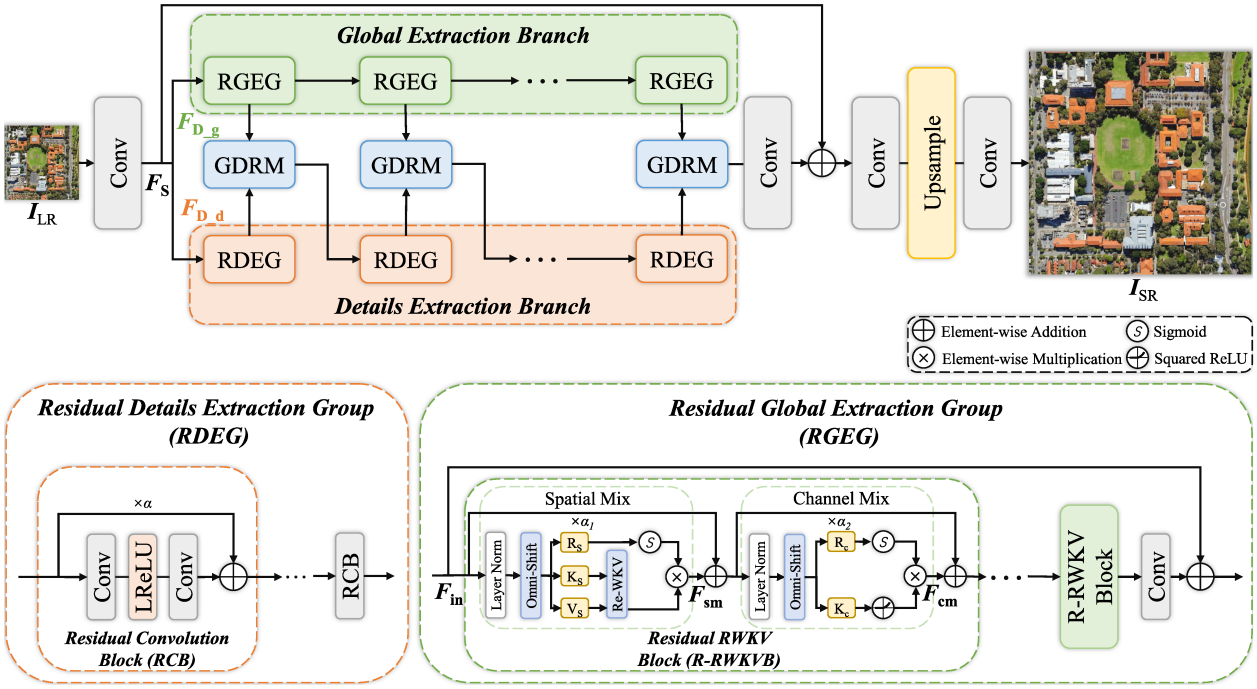

Fig. 3. Overview of the proposed GDSR.

C. Residual Global Extraction Group

Inspired by the recent success of RWKV in computer vision applications [20], [33], we seek to utilize its strengths in modeling long-range dependencies in RSIs. As illustrated in Fig. 3, the proposed RGEG consists of multiple residual RWKV blocks (R-RWKVBs). The spatial mix (sm) module is designed to establish long-range dependencies among tokens across the spatial dimension. Given an input feature flattened into a one-dimensional sequence $\boldsymbol{F}_{\text{in}} \in \mathbb{R}^{T\times C}$, where $T = H \times W$ represents the total number of tokens, the spatial mix module begins by applying layer normalization (LN) followed by the Omni-Shift operation [20]. The output is then passed through three parallel linear projection layers to obtain the receptance $R_s \in \mathbb{R}^{T\times C}$, key $K_s \in \mathbb{R}^{T\times C}$, and value $V_s \in \mathbb{R}^{T\times C}$

$$\boldsymbol{F}_s = \text{Omni-Shift}(\text{LN}(\boldsymbol{F}_{\text{in}})) \tag{4}$$

$$R_s = \boldsymbol{F}_s W_{R_s}, \quad K_s = \boldsymbol{F}_s W_{K_s}, \quad V_s = \boldsymbol{F}_s W_{V_s} \tag{5}$$

where W_{R_s}, W_{K_s}, and W_{V_s} are linear projection matrices. The key K_s and value V_s are then processed through the Re-WKV attention mechanism [20] to compute the global attention output $O_{\text{attn}} \in \mathbb{R}^{T\times C}$. Finally, the receptance R_s, after gating through a sigmoid function $\sigma(\cdot)$, modulates the attention output via element-wise multiplication

$$O_{\text{attn}} = \text{Re-WKV}(K_s, V_s) \tag{6}$$

$$\boldsymbol{F}_{\text{sm}} = (\sigma(R_s) \odot O_{\text{attn}})W_{\text{sm}} \tag{7}$$

where $\boldsymbol{F}_{\text{sm}}$ represents the spatial mix output, $\odot$ denotes element-wise multiplication, and W_{sm} is the output projection matrix. The channel mix (cm) module focuses on channel-wise feature fusion. Similar to the spatial mix module, it starts with LN and Omni-Shift, and the receptance $R_c \in \mathbb{R}^{T\times C}$, key $K_c \in \mathbb{R}^{T\times C}$, and value $V_c \in \mathbb{R}^{T\times C}$ are computed as follows:

$$\boldsymbol{F}_c = \text{Omni-Shift}(\text{LN}(\alpha_1 \boldsymbol{F}_{\text{in}} + \boldsymbol{F}_{\text{sm}})) \tag{8}$$

$$R_c = \boldsymbol{F}_c W_{R_c}, \quad K_c = \boldsymbol{F}_c W_{K_c}, \quad V_c = \gamma(K_c) W_{V_c} \tag{9}$$

where W_{R_c}, W_{K_c}, and W_{V_c} are linear projection matrices, and $\gamma(\cdot)$ is the squared ReLU activation function, which enhances nonlinearity. Notably, the transformation from $\boldsymbol{F}_c$ to K_c to V_c involves a multilayer perception consisting of W_{K_c}, $\gamma(\cdot)$, and W_{V_c}, facilitating channel-wise feature fusion. The final channel mix output $\boldsymbol{F}_{\text{cm}}$ is obtained through

$$\boldsymbol{F}_{\text{cm}} = (\sigma(R_c) \odot V_c)W_{\text{cm}} \tag{10}$$

where W_{cm} is the output projection matrix. Finally, RGEG can be formulated as

$$R = \alpha_2(\alpha_1 \boldsymbol{F}_{\text{in}} + \boldsymbol{F}_{\text{sm}}) + \boldsymbol{F}_{\text{cm}} \tag{11}$$

$$\text{RGEG}^k = WR^l(\cdots R^1(\text{RGEG}^{k-1})\cdots) + \text{RGEG}^{k-1} \tag{12}$$

where RGEG^k and RGEG^{k-1} denote the feature map of the kth RGEG and (k−1)th RGEG, respectively. R^1 is the lth R-RWKVB. W is a convolutional layer that serves to enhance the translational equivariance of the RWKV layer, while α_1 and α_2 are learnable scale factors.

D. Global-Detail Reconstruction Module

Remote sensing images often exhibit intricate spatial details and large-scale contextual information, necessitating a specialized module to harmonize these features for enhanced reconstruction performance. To effectively integrate the complementary features extracted by the RDEG and the RGEG, the GDRM is designed as a fusion mechanism tailored for RSI-SR tasks.

To harmonize and align these two feature representations, the GDRM employs a permuted spatial attention module (PSAM), which is designed to be computationally efficient and structurally simple. As illustrated in Fig. 4, the module begins

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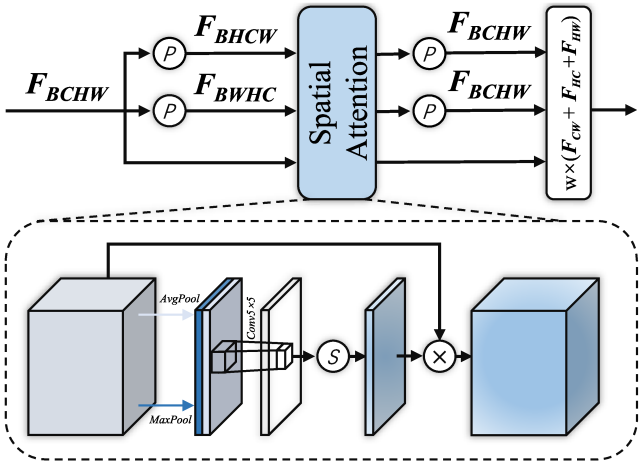

Fig. 4. Structure of PSAM.

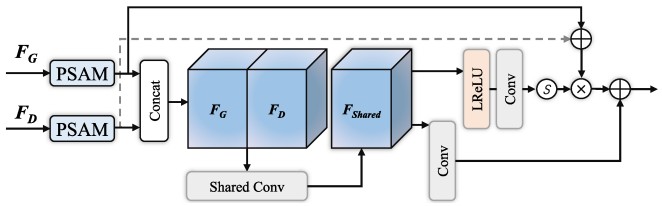

Fig. 5. Structure of GDRM.

by applying a permute operation to the input feature map, reorganizing it into three distinct formats, each capturing specific spatial relationships: height–width (HW), channel–width (CW), and height–channel (HC). Spatial attention [43] is subsequently applied to each permuted representation independently, enabling the modeling of dependencies along these dimensions. The processed features are then permuted back to their original format, summed together, and scaled by a coefficient. Formally, the aligned feature maps are computed as

$$\boldsymbol{F}_{\mathrm{CW}}, \boldsymbol{F}_{\mathrm{HC}} = \mathrm{Permute}(\mathrm{SA}(\mathrm{Permute}(\boldsymbol{F}_{\mathrm{in}})))$$
$$\boldsymbol{F}_{\mathrm{HW}} = \mathrm{SA}(\boldsymbol{F}_{\mathrm{in}}) \tag{13}$$
$$\boldsymbol{F}_{\mathrm{aligned}} = W(\boldsymbol{F}_{\mathrm{CW}} + \boldsymbol{F}_{\mathrm{HC}} + \boldsymbol{F}_{\mathrm{HW}}) \tag{14}$$

where $\mathrm{SA}(\cdot)$ denotes the spatial attention operation and W is a fixed coefficient. As illustrated in Fig. 5, the GDRM receives two input feature maps: one capturing local details $\boldsymbol{F}_D \in \mathbb{R}^{H\times W\times C}$ from the RDEG branch and another representing global contextual information $\boldsymbol{F}_G \in \mathbb{R}^{H\times W\times C}$ from the RGEG branch. The GDRM then processes these feature maps through a series of operations, which can be expressed as

$$\boldsymbol{F}_G = \mathrm{PSAM}(\boldsymbol{F}_G), \quad \boldsymbol{F}_D = \mathrm{PSAM}(\boldsymbol{F}_D) \tag{15}$$
$$\boldsymbol{F}_{\mathrm{shared}} = W_{\mathrm{shared}}(\mathrm{cat}(\boldsymbol{F}_G, \boldsymbol{F}_D)),$$
$$w = \sigma W_0(\mathrm{LReLU}(\boldsymbol{F}_{\mathrm{shared}})),$$
$$b = W_1(\boldsymbol{F}_{\mathrm{shared}}) \tag{16}$$
$$\boldsymbol{F}_{\mathrm{GDRM}} = w(\boldsymbol{F}_G + \boldsymbol{F}_D) + b. \tag{17}$$

E. Dual-Group Multiscale Wavelet Loss

The proposed wavelet loss construction follows the paradigm established in prior studies [40], [41]. As illustrated in Fig. 6, initially, both the HR image and SR image are transformed into the YCbCr color space. The stationary WT (SWT) is subsequently applied exclusively to the luminance (Y) channel, as human visual perception demonstrates higher sensitivity to structural details in this component. Through SWT decomposition, one low-frequency (LF) subband (LL) and three HF subbands (LH, HL, HH) are generated. The LL subband encapsulates coarse structural information, while the HF components encode directional details: LH captures horizontal features, HL preserves vertical characteristics, and HH retains diagonal patterns. These HF subbands are concatenated along the channel dimension to form a composite HF representation $\boldsymbol{H} \in \mathbb{R}^{H\times W\times 3}$, whereas the LL subband is retained as $\boldsymbol{L} \in \mathbb{R}^{H\times W\times 1}$.

TABLE I
GEOGRAPHICAL DISTRIBUTION OF COUNTIES IN THE RSSRD-QH DATASET

County	Longitude (°E)	Latitude (°N)
Maqin	100.219 to 100.285	34.264 to 34.7
Gande	100.047 to 100.719	34.054 to 34.204
Dari	98.7845 to 98.996	33.485 to 33.925
Maduo	97.929 to 98.129	34.668 to 34.698
Henan	101.374 to 101.759	34.307 to 34.707
Gangcha	100.406 to 100.509	37.351 to 37.671
Tianjun	98.409 to 98.624	37.45 to 37.4855

TABLE II
ABLATION STUDIES FOR THE NUMBER OF RCB AND R-RWKVB. THE CONFIGURATION WITH 12 RCBS AND 6 R-RWKVBS DENOTES THE PROPOSED GDSR MODEL. BOLD INDICATES THE MODEL PROPOSED IN THIS WORK

RCBs	R-RWKVBs	PSNR ↑	SSIM ↑	#Param.	FLOPs	FPS
1	6	28.96	0.7658	5.86M	151.23G	15.7
4		28.96	0.7657	7.85M	202.37G	15.2
8		29.00	0.7677	10.51M	270.55G	14.5
12		**29.03**	**0.7686**	**13.17M**	**338.73G**	**14.8**
16		29.01	0.7680	15.83M	406.92G	13.3
20		29.03	0.7688	18.48M	475.10G	12.6
12	1	29.00	0.7676	10.62M	272.97G	40.9
	2	29.01	0.7678	11.13M	286.12G	29.4
	4	29.01	0.7679	12.15M	312.43G	16.2
	8	29.01	0.7679	14.19M	365.04G	10.5

TABLE III
ABLATION STUDIES FOR THE GDRM. F2B (FEEDBACK TO BOTH RDEG AND RGEG) REFERS TO THE STRATEGY WHERE THE OUTPUT OF THE GDRM IS SENT BACK TO BOTH THE RDEG AND THE RGEG, ENABLING UPDATES TO THE RGEG FEATURES. F2D (FEEDBACK TO RDEG ONLY) REFERS TO THE STRATEGY WHERE THE OUTPUT OF THE GDRM IS SENT BACK ONLY TO THE RDEG, WHILE THE RGEG FEATURES CONTINUE TO PROPAGATE UNCHANGED FROM PREVIOUS LAYERS. BOLD INDICATES THE MODEL PROPOSED IN THIS WORK

Method	PSNR (dB) ↑	SSIM ↑	LPIPS ↓
RDEG Branch	28.94	0.7656	0.3554
RGEG Branch	28.94	0.7652	0.3519
w/o PSAM	29.01	0.7677	0.3485
F2B	29.01	0.7681	0.3476
F2D(GDSR)	**29.03**	**0.7686**	**0.3476**

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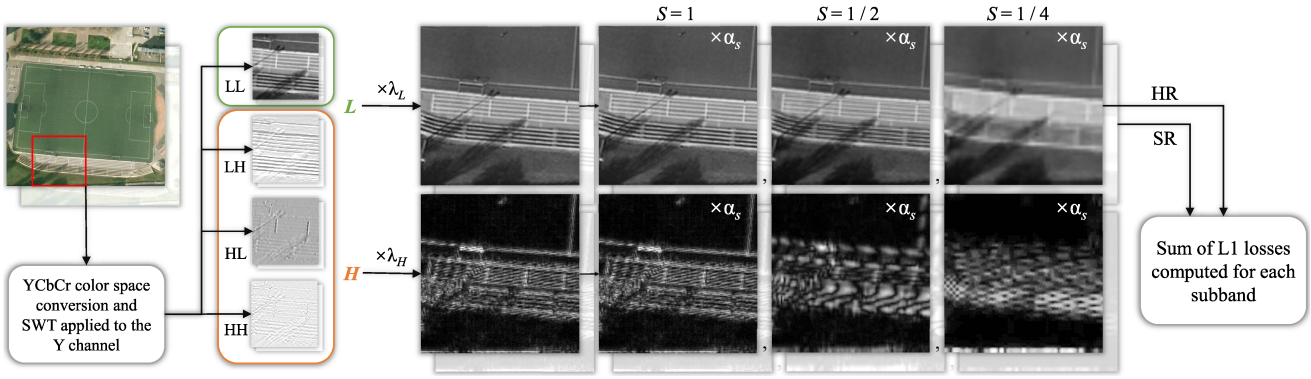

Fig. 6. Overview of the proposed dual-group multiscale wavelet loss. Zoomed-in view with brightness-optimized HF subbands for better observation.

TABLE IV
ABLATION STUDIES FOR THE RWKV

Method	PSNR/SSIM ↑	LPIPS ↓	#Param.	FLOPs	FPS
GDSR_TC	28.89/0.7633	0.3565	12.83M	337.26G	7.6
GDSR_MC	28.97/0.7663	0.3514	13.99M	350.35G	9.1
GDSR	**29.03/0.7686**	**0.3476**	**13.17M**	**338.73G**	**14.8**

The wavelet loss function is formulated as the $\mathcal{L}_1$ distance between the corresponding subbands of the generated SR image and the HR image. The total wavelet loss is expressed as

$$\mathcal{L}_{\text{wav}} = \lambda_L \sum_s \alpha_s \left\| \boldsymbol{L}^s_{\text{SR}} - \boldsymbol{L}^s_{\text{HR}} \right\|_1 + \lambda_H \sum_s \alpha_s \left\| \boldsymbol{H}^s_{\text{SR}} - \boldsymbol{H}^s_{\text{HR}} \right\|_1 \tag{18}$$

where $s \in \{1, (1/2), (1/4)\}$ denotes three spatial resolutions, $\boldsymbol{L}^s_{\text{SR}}$, $\boldsymbol{L}^s_{\text{HR}}$, $\boldsymbol{H}^s_{\text{SR}}$, and $\boldsymbol{H}^s_{\text{HR}}$ represent the LF/HF subbands of the SR/HR images at scale s, α_s are resolution-specific weights satisfying $\sum \alpha_s = 1$ (specific settings for α_s can be found in Section IV-C2, but generally, arbitrary α_s can be used), and λ_L and λ_H control the global contributions of LF/HF components. The overall loss for the training with the wavelet loss is given by

$$\mathcal{L}_{\text{rec}} = \|\boldsymbol{I}_{\text{HR}} - \boldsymbol{I}_{\text{SR}}\|_1 \tag{19}$$

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{wav}}. \tag{20}$$

IV. EXPERIMENT

A. Datasets and Evaluation

This article reports the performance of SR on three RSI datasets, including the publicly available AID [44], UCMerced [45], and the custom RSSRD-QH dataset.

The AID dataset consists of 30 categories, with the data in each category randomly divided into two equal parts for training and testing. Additionally, 20% of the training data was reserved as a validation set, resulting in 4000 training images, 1000 validation images, and 5000 test images. Similarly, the UCMerced dataset, comprising 21 categories with a total of 2100 images, was processed identically, yielding 945 training images, 105 validation images (10% of the training data), and 1050 test images.

The RSSRD-QH dataset is located in representative watershed units within the Sanjiangyuan region and the Qinghai Lake surrounding area. The geographical distribution of the counties covered by the dataset, including their longitude and latitude ranges, is summarized in Table I. The dataset was collected using uncrewed aerial vehicles, with flight altitudes carefully controlled between 30 and 50 m. The average spatial resolution of the captured imagery is 0.01 m, providing comprehensive coverage of various landscape elements across different elevation gradients, including terraces, floodplains, gentle slopes, and steep slopes within these watershed units. From each Watershed, we randomly selected 50% of the images for the training set and 50% for the test set, with 20% of the training set randomly selected for validation. This resulted in a training set of 2242 images, a validation set of 561 images, and a test set of 2804 images. The HR images in the AID and RSSRD-QH datasets were 600 × 600 pixels, while those in the UCMerced dataset were 256 × 256 pixels.

The proposed method, along with other competing methods, was evaluated on the test datasets using two classic full-reference metrics: the peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM) [46]. Additionally, the LPIPS [47] metric was used to capture the perceptual quality of the images. Notably, the PSNR and SSIM were calculated based on the luminance channel (Y) of the YCbCr color space, while LPIPS evaluated the perceptual similarity between images by extracting features from a pretrained deep network, specifically the AlexNet [48], which was employed in this study.

B. Degradation Model and Implementation Details

We employed both bicubic downsampling and a comprehensive degradation model (CDM) to simulate LR images during training. Based on existing blind SR methods [49], [50] and the characteristics of remote sensing sensors [51], [52], the CDM for LR image synthesis included isotropic Gaussian blur, motion blur, scaling with different interpolation methods, additive Gaussian noise, and JPEG compression noise, all applied in random order.

Our GDSR performed deep feature exploration through 4 RGEGs, RDEGs, and GDRMs. Each RGEG consisted of 6 R-RWKVBs, and each RDEG contains 12 RCBs. Empirically, we set the internal channel dimension to $c = 96$. For the GDSR_TC and GDSR_MC, the RGEG branches correspond

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TABLE V

ABLATION STUDIES FOR NETWORK DEPTH ON PERFORMANCE. BOLD INDICATES THE MODEL PROPOSED IN THIS WORK

Depth	PSNR/SSIM ↑	LPIPS ↓	#Param.	FLOPs	FPS
1	28.89/0.7633	0.3565	3.65M	94.06G	53.0
2	28.97/0.7663	0.3514	6.82M	175.62G	27.3
4	**29.03/0.7686**	**0.3476**	**13.17M**	**338.73G**	**14.8**
6	29.03/0.7686	0.3463	19.52M	501.85G	9.3
8	29.04/0.7692	0.3456	25.86M	664.96G	7.0
10	29.05/0.7695	0.3450	32.21M	828.07G	5.6

TABLE VI

ABLATION STUDIES FOR THE PROPOSED DUAL-GROUP MULTISCALE WAVELET LOSS. WE COMPARE OUR DUAL-GROUP MULTISCALE WAVELET LOSS WITH TWO BASELINE CONFIGURATIONS FROM [41]: 1) UNIFORM COEFFICIENT ALLOCATION (ALL LL/LH/HL/HH SUBBANDS WITH $\lambda = 0.05$) AND 2) ASYMMETRIC COEFFICIENTS (LL/HH $\lambda = 0.05$, LH/HL $\lambda = 0.01$) WITH ISOLATED SUBBAND PROCESSING. PROPOSED LOSS PARAMETERS: $\lambda_L = 0.05$, $\lambda_H = 0.01$, $\alpha = \{0.6, 0.3, 0.1\}$ FOR $s = \{1, (1/2), (1/4)\}$. ALL EVALUATED WAVELET-BASED METHODS EMPLOY THE "SYM19" WAVELET WITH 2-LEVEL DECOMPOSITION

Method	PSNR (dB) ↑	SSIM ↑	LPIPS ↓
$\mathcal{L}_{\text{rec}}$ loss	29.0329	0.7686	0.3476
Baseline [41]: Uniform coefficients	29.0182	0.7676	0.3565
Baseline [41]: Asymmetric coefficients	29.0413	0.7687	0.3534
Proposed: Dual-Group	29.0468	0.7690	0.3535
Proposed: Dual-Group Multi-Scale	**29.0514**	**0.7694**	**0.3519**

to the RTSB and ResidualGroup of SwinIR [12] and MambaIR [15], respectively, with all other settings remaining consistent with GDSR.

During training, 64 × 64 patches were randomly cropped from LR RSIs, along with their HR counterparts scaled by the target scale factor (×2, ×3, and ×4). The batch size was set to 16. The initial learning rate was set to 1×10^{-4} and optimized using the Adam optimizer, where $\beta 1 = 0.9$ and $\beta 2 = 0.99$, with a step learning rate scheduler halving the rate at the 100th epoch over a total of 200 training epochs. All SR models were implemented in the PyTorch framework. The FLOPs results were calculated using an input tensor of size 1 × 3 × 160 × 160, and the computations were performed using torch-operation-counter.[1] The inference times were tested on 300 random images with a size of 3 × 160 × 160. All experiments were conducted on an NVIDIA A800 80GB GPU.

C. Ablation Study

In this section, we discuss the proposed GDSR in depth by investigating the effect of its major components and their variants. All experiments were conducted under the ×3 super-resolution (SR) configuration with the $\mathcal{L}_{\text{rec}}$ loss function in the AID [44] dataset.

1) Effect of Key Modules:

a) Effect of RDEG and RGEG: Ablation studies were conducted to evaluate the impact of varying RCBs in RDEG and R-RWKVBs in RGEG on model performance. As shown in Tables II and III, the baseline model using only the RGEG branch yielded a PSNR of 28.94 dB and SSIM of 0.7652. Introducing a single RCB to RDEG improved performance to 28.96 dB PSNR and 0.7658 SSIM, validating the RCB's ability to enhance reconstruction quality. Scaling the RCB count to 12 further increased PSNR to 29.03 dB and SSIM to 0.7686, with a computational footprint of 13.17M parameters and 338.73G FLOPs. However, beyond 12 RCBs, performance plateaued (e.g., 29.01 dB with 16 RCBs) or exhibited instability (e.g., 29.03 dB with 20 RCBs), while computational costs grew linearly, reducing FPS from 14.8 to 12.6. This suggests diminishing returns from excessive RCB stacking, necessitating a balance between accuracy and efficiency.

Similarly, experiments on RGEG's R-RWKVBs revealed analogous trends. Fixing RCBs at 12 and increasing R-RWKVBs from 1 to 8 marginally improved the PSNR and SSIM but incurred significant computational overhead, with parameters rising from 10.62M to 14.19M and FPS dropping sharply from 40.9 to 10.5. Notably, the joint configuration of 12 RCBs and 6 R-RWKVBs achieved optimal trade-offs, outperforming isolated branch evaluations (as shown in Table III). These results underscore the complementary roles of RDEG and RGEG in feature refinement while emphasizing the necessity of constrained component counts to mitigate redundancy and computational bloat in practical deployments.

b) Effect of GDRM: Ablation experiments in Table III demonstrated that the dual-branch GDSR model outperformed single-branch alternatives, achieving superior PSNR, SSIM, and LPIPS values. This performance gain was attributed to the GDRM, which systematically integrated dual-branch features through its PSAM and adaptive fusion mechanism (16 and 17). From a spatial–frequency duality perspective, RGEG's global attention mechanisms implicitly exhibited low-pass filtering characteristics to maintain structural continuity, while RDEG's convolutional operations inherently emphasized HF components for local detail enhancement [21]. As visualized in Fig. 7, the dual-branch structure of GDSR demonstrated complementary feature specialization with GDRM dynamically fusing both to enable efficient reconstruction.

Visualizations in Fig. 8 revealed that RGEG activated distributed pixels across broader regions, whereas RDEG focused

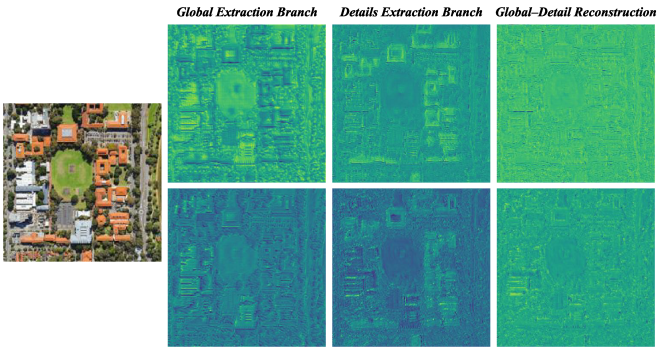

Fig. 7. Feature visualization comparisons. First row: channel-0 feature maps from the PSAM-processed final outputs of RGEG and RDEG. Second row: channel-1 feature maps from the same modules. Visualizations of GDRM outputs at corresponding network stages are provided in parallel for comparison.

[1] https://github.com/SamirMoustafa/torch-operation-counter

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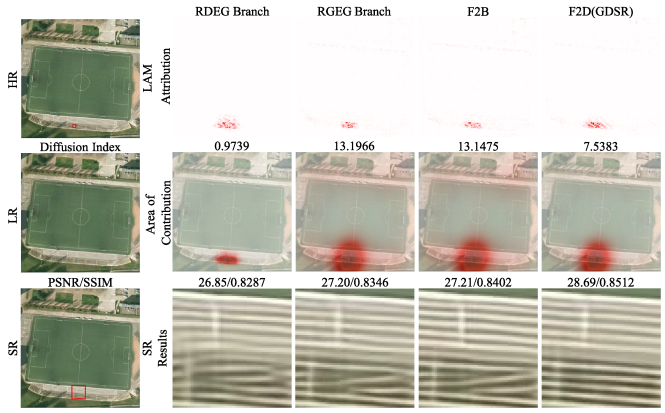

Fig. 8. Visualization of LAMs [53]. The LAM maps represent the importance of each pixel in the input LR image with respect to the SR of the patch marked with a red box. The DI [53] reflects the range of involved pixels. A higher DI represents a wider range of attention.

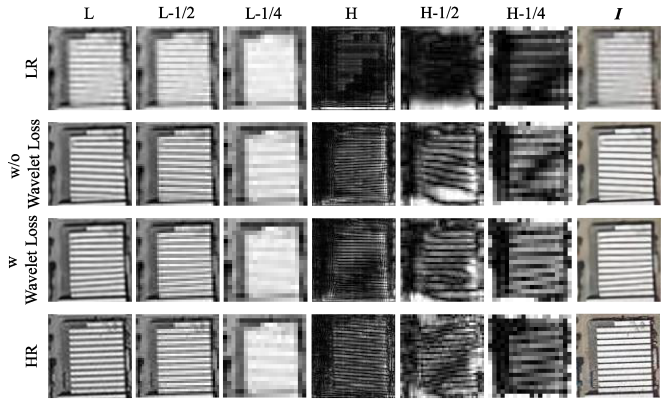

Fig. 9. Visualization of dual-group multiscale wavelet loss effectiveness in GDSR. Brightness-optimized HF subbands for better observation.

on local patterns. The F2D strategy further enhanced this synergy by routing refined features exclusively to RDEG, prioritizing localized refinement over redundant global propagation. This design addressed a critical theoretical limitation: while RGEG aggregated broader pixel information, effective reconstruction required targeted utilization of spatial relationships within receptive fields. F2D's spectral separation strategy specifically resolved conflicts observed in F2B, where dual-path feedback introduced frequency interference, as global attention diluted HF details while convolutions struggled to recover them.

The quantitative results confirmed that GDSR's dual-branch framework, coupled with GDRM's reconstruction, systematically unified heterogeneous features to achieve both perceptual and pixel-wise accuracy. This validated the theoretical rationale that explicitly decoupling global (low-frequency) and local (HF) processing paths, followed by attention-driven fusion, was essential for overcoming the limitations of conventional single-path architectures.

c) Effect of RWKV: To investigate the effect of RWKV, ablation experiments were conducted by replacing RWKV with Transformer [12] (GDSR_TC) and Mamba [15] (GDSR_MC) while keeping other components unchanged. As shown in Table IV, the RWKV-based GDSR achieves the best performance in terms of the PSNR, SSIM, and LPIPS, outperforming both Transformer and Mamba variants. Notably, RWKV demonstrates superior computational efficiency with 14.8 FPS, significantly exceeding GDSR_TC and GDSR_MC. While maintaining moderate parameter size and FLOPs, RWKV achieves an optimal balance between reconstruction quality and computational cost. This validates RWKV's effectiveness in capturing long-range dependencies while mitigating the quadratic complexity limitations of conventional attention mechanisms.

d) Effect of Network Depth on Performance: To investigate the effect of network depth on the performance of GDSR, we varied the shared depth of both the RDEG and RGEG branches while keeping other design parameters constant. As illustrated in Table V, the experimental outcomes consistently demonstrated an enhancement in performance with deeper networks. Moreover, an evaluation of the tradeoffs in terms of computational complexity was conducted. As the network depth increased, the model's parameter count, FLOPs, and FPS were also scaled accordingly. These results supported the effectiveness of GDSR's dual-branch design, where the shared depth allowed for both high-quality SR and manageable computational overhead.

2) Effect of Wavelet Loss: Ablation experiments were conducted across multiple configurations to evaluate the effectiveness of incorporating wavelet loss in training. Quantitative results (Table VI) demonstrate that our full configuration achieves the highest PSNR and SSIM while attaining the lowest LPIPS, outperforming both baselines and the dual-group-only variant. Visual evidence in Fig. 9 further corroborates these findings; the inclusion of the proposed dual-group multiscale wavelet loss significantly improved the reconstruction, particularly in the HF components. This improvement is attributed to the synergistic integration of dual-coefficient grouping strategy and cross-resolution frequency consistency, which collectively overcome the rigidity of fixed-scale wavelet methods that process subbands in isolation. These results highlight wavelet loss's role in guiding the model to recover intricate structures and accurately reconstruct complex textures, thereby enhancing overall image quality.

However, as evidenced in Table VI, all wavelet loss variants exhibit higher LPIPS values compared to the $\mathcal{L}_{\text{rec}}$ baseline, indicating an inherent challenge in reconciling frequency-domain fidelity with perceptual authenticity. Specifically, the uniform coefficient configuration [41] yields the worst LPIPS due to its undifferentiated weighting. Our dual-group multiscale design strategically mitigates this tradeoff. While the proposed method's LPIPS remains marginally higher than $\mathcal{L}_{\text{rec}}$, it achieves the lowest LPIPS among wavelet-based approaches. The resultant configuration delivers superior PSNR/SSIM with minimal perceptual compromise, demonstrating that while frequency-domain constraints inherently elevate LPIPS, our method optimally balances this tradeoff with substantial PSNR/SSIM gains.

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TABLE VII

QUANTITATIVE COMPARISON ON AID [44] AND UCMERCED [45] TEST SET IN TERMS OF THE PSNR, SSIM, AND LPIPS, WHERE THE BEST PERFORMANCE IS HIGHLIGHTED IN RED

Methods	Type	AID x2			AID x3			AID x4		
		PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓
Bicubic	Bicubic Synthetic	30.48	0.8269	0.2769	27.89	0.7238	0.4400	26.43	0.6541	0.5640
SRCNN [6]		30.87	0.8352	0.2711	28.22	0.7363	0.3973	26.68	0.6679	0.4862
EDSR [9]		31.70	0.8560	0.2416	28.94	0.7653	0.3513	27.30	0.6994	0.4381
SRGAN [8]		31.57	0.8523	0.2481	28.80	0.7597	0.3601	27.20	0.6939	0.4466
SwinIR [12]		31.71	0.8567	0.2370	28.95	0.7658	0.3512	27.32	0.6993	0.4373
HAT [13]		31.77	0.8579	0.2351	28.99	0.7671	0.3479	27.34	0.7004	0.4342
HAUNet [25]		31.48	0.8507	0.2503	28.75	0.7580	0.3657	27.15	0.6913	0.4523
SPT [27]		31.61	0.8547	0.2391	28.86	0.7620	0.3553	27.26	0.6962	0.4418
ConvFormerSR [28]		31.64	0.8551	0.2298	28.90	0.7637	0.3516	27.26	0.6964	0.4396
MambaIR [15]		31.78	0.8583	0.2354	29.02	0.7684	0.3487	27.37	0.7019	0.4348
FreMamba [16]		31.62	0.8558	**0.2215**	28.96	0.7672	**0.3410**	27.32	0.7010	**0.4255**
GDSR (Ours)		**31.81**	**0.8592**	0.2366	**29.05**	**0.7694**	0.3519	**27.39**	**0.7022**	0.4376
Bicubic	CDM Synthetic	24.11	0.5545	0.6046	23.15	0.5238	0.6911	22.54	0.5091	0.7543
SRCNN [6]		24.44	0.5854	0.5836	23.38	0.5455	0.6569	22.70	0.5245	0.7100
EDSR [9]		24.90	0.6153	0.5341	23.91	0.5698	0.6139	23.16	0.5444	0.6753
SRGAN [8]		24.78	0.6070	0.5609	23.78	0.5648	0.6314	23.02	0.5396	0.6909
SwinIR [12]		24.95	0.6144	0.5353	23.89	0.5705	0.6169	23.17	0.5452	0.6737
HAT [13]		25.06	0.6183	0.5276	23.99	0.5737	0.6080	23.26	0.5475	0.6671
HAUNet [25]		25.03	0.6169	0.5287	23.90	0.5692	0.6185	23.19	0.5441	0.6783
SPT [27]		24.86	0.6086	0.5457	23.88	0.5679	0.6257	23.20	0.5447	0.6749
ConvFormerSR [28]		25.04	0.6137	0.5378	23.97	0.5710	0.6118	23.23	0.5448	0.6742
MambaIR [15]		25.17	0.6230	0.5120	24.07	0.5763	**0.6000**	23.32	0.5497	**0.6590**
FreMamba [16]		25.00	0.6227	**0.4977**	23.89	0.5692	0.6148	23.06	0.5403	0.6735
GDSR (Ours)		**25.28**	**0.6256**	0.5225	**24.12**	**0.5771**	0.6106	**23.39**	**0.5510**	0.6664
		UCMerced x2			UCMerced x3			UCMerced x4		
Bicubic	Bicubic Synthetic	31.10	0.8892	0.2013	27.59	0.7731	0.3548	25.73	0.6813	0.4838
SRCNN [6]		32.33	0.9102	0.1635	28.48	0.8028	0.2919	26.39	0.7137	0.3944
EDSR [9]		34.31	0.9337	0.0951	29.98	0.8464	0.2048	27.70	0.7644	0.3002
SRGAN [8]		34.07	0.9317	0.1007	29.79	0.8427	0.2207	27.53	0.7571	0.3181
SwinIR [12]		34.35	0.9344	0.0942	30.06	0.8514	0.2018	27.75	0.7684	0.2956
HAT [13]		34.30	0.9339	0.0936	30.08	**0.8527**	0.1979	27.77	**0.7700**	0.2914
HAUNet [25]		33.68	0.9270	0.1113	29.47	0.8343	0.2333	27.25	0.7479	0.3381
SPT [27]		34.12	0.9319	0.0968	29.79	0.8439	0.2115	27.55	0.7613	0.3072
ConvFormerSR [28]		34.27	0.9330	0.0945	29.89	0.8461	0.2123	27.59	0.7614	0.3109
MambaIR [15]		34.52	**0.9356**	**0.0898**	30.14	0.8513	**0.1934**	27.75	0.7694	**0.2874**
FreMamba [16]		33.87	0.9283	0.1053	29.95	0.8471	0.2025	27.61	0.7639	0.2992
GDSR (Ours)		**34.53**	0.9354	0.0918	**30.15**	0.8500	0.2057	**27.79**	0.7683	0.3099
Bicubic	CDM Synthetic	22.46	0.5128	0.6323	21.35	0.4678	0.7330	20.86	0.4558	0.7846
SRCNN [6]		22.71	0.5392	0.6277	21.48	0.4860	0.7077	20.96	0.4702	0.7593
EDSR [9]		22.99	0.5635	0.5485	21.67	0.5056	0.6402	21.10	0.4850	0.6917
SRGAN [8]		22.96	0.5603	0.5675	21.65	0.5041	0.6569	21.10	0.4836	0.7051
SwinIR [12]		23.09	0.5682	0.5496	21.77	0.5096	0.6481	21.18	0.4882	0.6900
HAT [13]		23.22	0.5758	0.5338	21.82	0.5129	0.6374	21.18	0.4895	0.6851
HAUNet [25]		23.03	0.5620	0.5635	21.69	0.5033	0.6655	21.07	0.4824	0.7117
SPT [27]		23.22	0.5724	0.5464	21.82	0.5083	0.6565	21.26	0.4884	0.7055
ConvFormerSR [28]		23.30	0.5739	0.5453	21.91	0.5124	0.6459	**21.27**	0.4898	0.6951
MambaIR [15]		23.23	0.5778	**0.5254**	21.76	0.5129	0.6247	21.16	**0.4903**	0.6749
FreMamba [16]		23.41	0.5783	0.5287	21.73	0.5129	**0.6200**	21.12	0.4894	**0.6723**
GDSR (Ours)		**23.45**	**0.5836**	0.5304	**21.93**	**0.5156**	0.6323	21.23	0.4894	0.6828

D. Comparisons With State-of-the-Art

1) Comparative Methods: CNN-based models, including SRCNN [6], EDSR [9], and HAUNet [25]; GAN-based models, including SRGAN [8]; Transformer-based models, including SwinIR [12], HAT [13], and SPT [27]; CNN–Transformer hybrid models, including ConvFormerSR [28]; and Mamba-based models, including MambaIR [15] and FreMamba [16], were involved in evaluating the SR performance of our GDSR against SOTA methods on RSI-SR.

2) Quantitative Evaluations: The quantitative evaluation results across different datasets and scaling factors were

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TABLE VIII
QUANTITATIVE COMPARISON ON THE RSSRD-QH TEST SET IN TERMS OF THE PSNR, SSIM, AND LPIPS, WHERE THE BEST PERFORMANCE IS HIGHLIGHTED IN RED

Methods	Type	RSSRD-QH x2			RSSRD-QH x3			RSSRD-QH x4		
		PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓
Bicubic	Bicubic Synthetic	29.37	0.8149	**0.2564**	26.45	0.6447	0.4243	24.92	0.5224	0.5764
SRCNN [6]		29.58	0.8218	0.3028	26.54	0.6512	0.4241	24.98	0.5268	0.5613
EDSR [9]		29.83	0.8299	0.2838	26.70	0.6598	0.4141	25.10	0.5343	0.5487
SRGAN [8]		29.80	0.8293	0.2859	26.68	0.6586	0.4170	25.08	0.5333	0.5522
SwinIR [12]		29.83	0.8305	0.2844	26.70	0.6597	0.4133	25.10	0.5349	0.5474
HAT [13]		29.84	0.8307	0.2836	26.71	0.6602	0.4111	25.10	0.5352	0.5454
HAUNet [25]		29.79	0.8289	0.2888	26.65	0.6569	0.4177	25.07	0.5326	0.5494
SPT [27]		29.83	0.8306	0.2875	26.69	0.6602	0.4161	25.10	0.5349	0.5478
ConvFormerSR [28]		29.82	0.8310	0.2786	26.69	0.6603	**0.4090**	25.10	0.5354	0.5452
MambaIR [15]		29.86	**0.8311**	0.2825	26.72	**0.6607**	0.4124	25.11	0.5359	0.5445
FreMamba [16]		29.80	0.8300	0.2811	26.70	0.6601	0.4112	25.11	**0.5366**	**0.5414**
GDSR (Ours)		**29.87**	0.8310	0.2823	**26.73**	0.6601	0.4130	**25.12**	0.5354	0.5465
Bicubic	CDM Synthetic	23.10	0.3991	**0.5996**	22.34	0.3525	**0.6993**	21.85	0.3284	**0.7764**
SRCNN [6]		23.27	0.4159	0.6520	22.44	0.3588	0.7523	21.91	0.3311	0.8078
EDSR [9]		23.67	0.4485	0.6269	22.74	0.3817	0.7283	22.15	0.3444	0.8028
SRGAN [8]		23.57	0.4387	0.6442	22.64	0.3759	0.7421	22.09	0.3408	0.8174
SwinIR [12]		23.68	0.4463	0.6232	22.72	0.3780	0.7432	22.15	0.3440	0.8104
HAT [13]		23.75	0.4528	0.6120	22.79	0.3823	0.7312	22.18	0.3456	0.8061
HAUNet [25]		23.72	0.4501	0.6136	22.76	0.3797	0.7360	22.16	0.3442	0.8096
SPT [27]		23.58	0.4353	0.6512	22.73	0.3743	0.7520	22.14	0.3415	0.8144
ConvFormerSR [28]		23.67	0.4398	0.6267	22.77	0.3779	0.7365	22.18	0.3436	0.8051
MambaIR [15]		**23.82**	**0.4568**	0.6036	22.84	0.3835	0.7387	22.23	0.3476	0.8003
FreMamba [16]		23.59	0.4329	0.6009	22.60	0.3660	0.7242	22.06	0.3358	0.7894
GDSR (Ours)		23.75	0.4454	0.6197	**22.87**	**0.3864**	0.7308	**22.27**	**0.3488**	0.8044

comprehensively analyzed through Tables VII and VIII to assess reconstruction accuracy and perceptual quality. On the AID dataset under bicubic degradation, the proposed GDSR demonstrated state-of-the-art performance with PSNR values of 31.81 dB (×2), 29.05 dB (×3), and 27.39 dB (×4), showing improvements of 0.04–0.11 dB over Transformer-based HAT and CNN-based EDSR. The structural preservation capability was further verified by SSIM metrics, where GDSR achieved 0.8592 (×2), 0.7694 (×3), and 0.7022 (×4). Although FreMamba exhibited superior LPIPS scores across ×2–4 scales, GDSR maintained competitive perceptual quality while balancing pixel-wise accuracy and visual realism. For more challenging CDM-synthesized data, GDSR showed remarkable robustness with consistent performance gains, surpassing MambaIR by 0.11-dB PSNR and 0.26% SSIM in ×2 SR. The superiority became more evident on the UCMerced dataset, where GDSR established new benchmarks in 7 out of 12 metrics. Notably, it achieved 0.17-dB PSNR improvement over MambaIR and 0.11-dB gain against HAT in ×3 SR under CDM degradation. Evaluation on the real-world RSSRD-QH dataset further validated the practical effectiveness of GDSR. Under bicubic degradation, the method attained peak PSNR values of 29.87 dB (×2), 26.73 dB (×3), and 25.12 dB (×4). In CDM degradation scenarios, GDSR exhibited particularly strong performance at high SR factors, reaching 22.87-dB PSNR (×3) and 22.27-dB PSNR (×4) with corresponding SSIM values of 0.3864 and 0.3488.

To further evaluate the generalization capability of the SR models across diverse remote sensing scenarios, we report the PSNR and SSIM results across 30 scene categories in the AID dataset under bicubic degradation, as shown in Table IX. As evidenced, GDSR exhibited stronger generalization capability compared to state-of-the-art methods, achieving the best performance in nearly all remote sensing scenarios. Specifically, for the "parking" and "playground" scenes, GDSR outperformed HAT by 0.13 and 0.1 dB in the PSNR and by 0.33% and 0.28% in the SSIM, respectively. Additionally, in the most complex categories, such as "church" and "dense residential," GDSR also achieved the best performance.

Moreover, as the receptive field of the models increased, performance trended upward for CNN-based EDSR, Transformer-based SwinIR, and Mamba-based MambaIR. This aligned with our experimental results on the RGEG and RDEG branches, where larger receptive fields enabled the activation of more pixels for image reconstruction, enhancing performance. However, the results from MambaIR and FreMamba, along with the experiments on F2B and F2C, indicated that spatial global modeling in complex RSIs had reached a bottleneck. Simply increasing the receptive field no longer yielded significant performance gains. A more practical solution was to effectively utilize large receptive fields to activate the appropriate regions and pixels for accurate image reconstruction. The outstanding generalization capability and superior performance of GDSR align with our motivation to address the challenges of large-scale RSIs efficiently through the introduction of a global-detail dual-branch structure.

3) Complexity and Efficiency Evaluation: To evaluate the complexity and computational efficiency of our proposed

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TABLE IX

QUANTITATIVE COMPARISON WITH SOTA SR METHODS ACROSS 30 SCENE CATEGORIES ON AID [44], WHERE THE BEST AND SECOND-BEST PSNR/SSIM PERFORMANCE ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Categoties	Bicubic		SRGAN [8]		EDSR [9]		HAT [13]		MambaIR [15]		ConvFormerSR [28]		FreMamba [16]		**GDSR (Ours)**	
	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑
Airport	27.43	0.7387	28.48	0.7765	28.63	0.7823	28.69	0.7837	**28.72**	**0.7856**	28.59	0.7805	28.68	0.7851	**28.74**	**0.7865**
Bare Land	34.45	0.8344	34.84	0.8422	34.92	0.8439	34.94	0.8445	**34.96**	**0.8449**	34.91	0.8436	34.93	0.8444	**34.98**	**0.8455**
Baseball Field	29.19	0.7786	30.27	0.8094	30.42	0.8137	30.45	0.8144	**30.52**	**0.8160**	30.35	0.8123	30.43	0.8147	**30.55**	**0.8169**
Beach	31.32	0.7947	31.86	0.8082	31.94	0.8103	31.99	0.8117	**32.00**	**0.8118**	31.94	0.8096	31.95	0.8113	**32.03**	**0.8124**
Bridge	30.19	0.8063	31.27	0.8312	31.46	0.8353	31.52	0.8363	**31.56**	**0.8372**	31.42	0.8339	31.52	0.8368	**31.59**	**0.8378**
Center	25.76	0.6980	27.18	0.7590	27.38	0.7665	27.47	0.7694	**27.51**	**0.7706**	27.33	0.7645	27.43	0.7692	**27.52**	**0.7714**
Church	23.09	0.6084	24.41	0.6807	24.56	0.6895	24.61	0.6920	**24.65**	**0.6941**	24.52	0.6875	24.56	0.6919	**24.68**	**0.6954**
Commercial	26.72	0.7178	27.59	0.7582	27.73	0.7645	27.77	0.7660	**27.81**	**0.7679**	27.68	0.7623	27.72	0.7659	**27.83**	**0.7692**
D-Residential	23.45	0.6301	24.46	0.6922	24.60	0.7004	24.64	0.7026	**24.69**	**0.7051**	24.56	0.6985	24.60	0.7029	**24.72**	**0.7067**
Desert	37.14	0.8869	37.54	0.8933	37.58	0.8944	**37.63**	**0.8951**	37.62	0.8949	37.58	0.8941	37.60	0.8946	**37.65**	**0.8954**
Farmland	32.32	0.7934	33.18	0.8166	33.33	0.8212	33.35	0.8216	**33.42**	**0.8236**	33.30	0.8202	33.39	0.8235	**33.45**	**0.8245**
Forest	27.67	0.6411	27.84	0.6514	27.90	0.6558	**27.95**	**0.6605**	27.95	0.6604	27.88	0.6552	27.92	0.6598	**27.99**	**0.6617**
Industrial	26.29	0.7063	27.43	0.7566	27.62	0.7647	27.67	0.7669	**27.71**	**0.7689**	27.56	0.7621	27.62	0.7665	**27.74**	**0.7701**
Meadow	32.07	0.7100	32.33	0.7136	32.38	0.7157	32.40	0.7167	**32.41**	0.7172	32.37	0.7154	32.39	**0.7175**	**32.43**	**0.7176**
M-Residential	26.32	0.6578	27.40	0.7055	27.55	0.7124	27.58	0.7136	**27.63**	**0.7159**	27.52	0.7113	27.55	0.7140	**27.66**	**0.7170**
Mountain	28.05	0.6978	28.41	0.7127	28.48	0.7163	**28.50**	**0.7184**	28.50	0.7183	28.46	0.7152	28.45	0.7171	**28.52**	**0.7192**
Park	27.14	0.7060	27.77	0.7332	27.87	0.7383	27.91	0.7406	**27.93**	**0.7415**	27.83	0.7365	27.86	0.7395	**27.96**	**0.7431**
Parking	24.11	0.7310	26.18	0.8059	26.51	0.8159	26.68	0.8205	**26.76**	0.8225	26.46	0.8145	26.70	**0.8226**	**26.81**	**0.8238**
Playground	29.33	0.7723	30.70	0.8132	30.94	0.8200	31.03	0.8216	**31.10**	**0.8235**	30.88	0.8182	31.03	0.8230	**31.13**	**0.8244**
Pond	28.76	0.7559	29.38	0.7749	29.46	0.7781	29.49	0.7790	**29.51**	**0.7796**	29.44	0.7769	29.46	0.7788	**29.53**	**0.7802**
Port	25.83	0.7727	26.94	0.8207	27.11	0.8255	27.18	0.8273	**27.21**	**0.8283**	27.05	0.8237	27.13	0.8267	**27.22**	**0.8280**
Railway Station	27.00	0.6965	27.87	0.7347	28.04	0.7434	28.11	0.7456	**28.13**	**0.7474**	27.99	0.7404	28.07	0.7461	**28.16**	**0.7489**
Resort	26.78	0.7160	27.67	0.7547	27.81	0.7601	27.86	0.7622	**27.89**	**0.7634**	27.78	0.7588	27.82	0.7619	**27.92**	**0.7644**
River	29.29	0.7201	29.74	0.7378	29.81	0.7412	29.84	0.7426	**29.85**	**0.7430**	29.79	0.7401	29.80	0.7422	**29.88**	**0.7439**
School	25.81	0.7046	26.81	0.7503	26.96	0.7568	27.00	0.7587	**27.04**	**0.7604**	26.91	0.7546	26.94	0.7577	**27.07**	**0.7619**
S-Residential	25.71	0.5790	26.15	0.6001	26.22	0.6055	26.25	0.6070	**26.27**	**0.6079**	26.21	0.6046	26.23	0.6075	**26.29**	**0.6080**
Square	27.51	0.7286	28.71	0.7782	28.90	0.7847	28.96	0.7870	**28.99**	**0.7881**	28.85	0.7831	28.92	0.7868	**29.03**	**0.7891**
Stadium	26.23	0.7241	27.55	0.7771	27.75	0.7845	27.82	0.7873	**27.85**	**0.7882**	27.68	0.7823	27.78	0.7867	**27.87**	**0.7889**
Storage Tanks	25.15	0.6667	26.25	0.7190	26.39	0.7255	26.41	0.7264	**26.45**	**0.7287**	26.33	0.7233	26.39	0.7274	**26.47**	**0.7298**
Viaduct	26.77	0.6817	27.70	0.7225	27.88	0.7310	27.91	0.7325	**27.98**	**0.7361**	27.81	0.7277	27.87	0.7326	**28.00**	**0.7378**
Average	27.89	0.7238	28.80	0.7597	28.94	0.7653	28.99	0.7671	**29.02**	**0.7684**	28.90	0.7637	28.96	0.7672	**29.05**	**0.7694**

TABLE X

COMPARISONS OF MODEL COMPLEXITY AND EFFICIENCY

Model	#Param.	FLOPs	Memory	FPS
Bicubic	-	-	22MB	27498.0
SRCNN [6]	0.02M	4.65G	140MB	1109.0
SRGAN [8]	1.55M	47.86G	300MB	323.3
EDSR [9]	43.68M	1119.98G	1068MB	35.1
SwinIR [12]	16.62M	446.67G	516MB	5.5
HAT [13]	20.81M	652.17G	1446MB	4.6
HAUNet [25]	2.64M	1353.07G	7658MB	3.1
SPT [27]	3.25M	668.46G	9974MB	2.2
ConvFormerSR [28]	16.49M	442.18G	500MB	7.0
MambaIR [15]	15.11M	372.85G	556MB	6.4
FreMamba [16]	11.46M	229.50G	546MB	5.5
GDSR (Ours)	13.17M	338.73G	518MB	14.8

models, we conducted quantitative assessments across various metrics. As shown in Table X, the bicubic interpolation technique inherently avoids the need for parameter optimization, resulting in the fastest inference speed within our evaluation scope. The CNN-based model achieved the highest inference speed among all tested deep learning models, primarily due to the inherent efficiency of convolutional parallel operations. The EDSR model, with its deep residual structure, had the largest number of parameters and FLOPs, yet achieved inference speeds several times faster than other architectures.

Transformer-based models, such as SwinIR and HAT, exhibited slower inference speeds compared to CNN-based models but achieved significant performance gains. Models based on Mamba, including MambaIR and FreMamba, leveraged their linear complexity and optimized inference processes, delivering superior performance along with relatively faster inference speeds. ConvFormerSR, which integrated Swin Transformer with CNN architectures, achieved faster inference speeds while maintaining competitive performance. GDSR, by incorporating RWKV with CNN, not only reduced the parameter count but also achieved inference speeds several times faster than Transformer- and Mamba-based models, while delivering enhanced performance. As demonstrated in Tables II–V, these results underscore the superiority of the global-detail dual-branch architecture design.

Comparison of Tables VII and X reveals that the complexity of our proposed model is comparable to most state-of-the-art models, while its efficiency is superior, and its performance is significantly better. These results highlight the practical potential of our model, demonstrating an effective balance between computational efficiency and performance, making it well-suited for real-world applications.

4) Qualitative Results: Visual comparisons on AID, AID_CDM, and RSSRD-QH_CDM with a scale factor of $\times 3$ are presented in Figs. 10–12. From these visual results, it was evident that our proposed GDSR was capable of restoring sharp edges and showcasing richer textures, particularly in capturing critical HF details in remote sensing images. For instance, when comparing the reconstructed "playground_161" in Fig. 10, the closest

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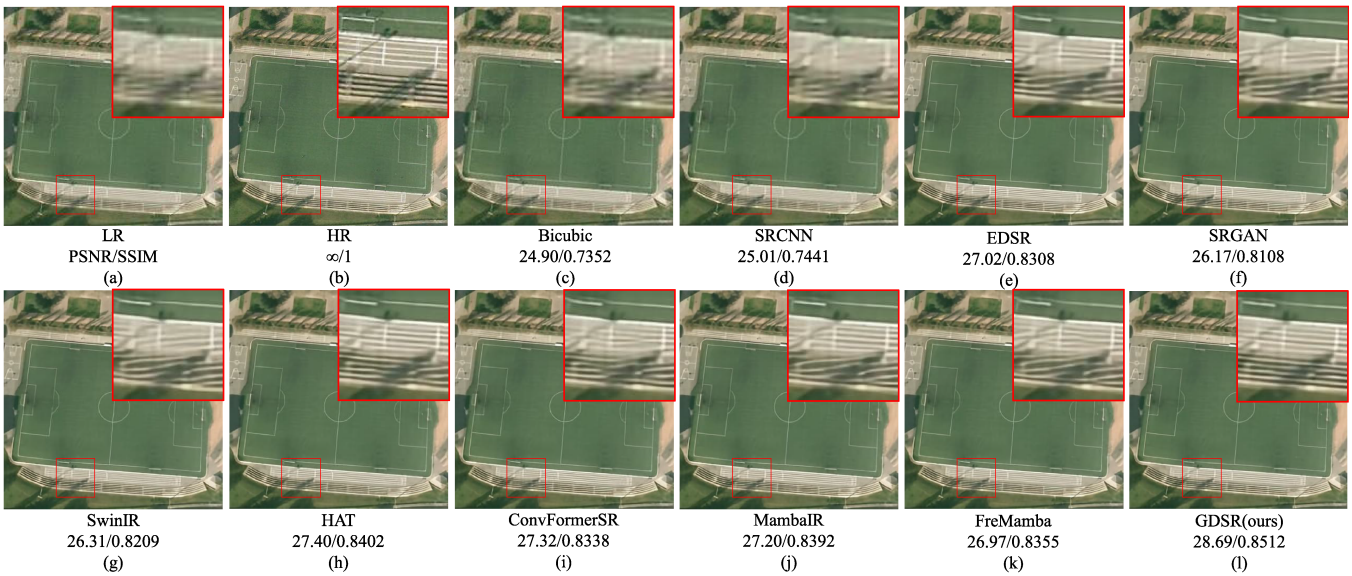

Fig. 10. Visual comparisons of our GDSR with CNN-, GAN-, Transformer-, and Mamba-based methods on the AID dataset's "playground_161" image with scale ×3. Zoomed-in view for better observation.

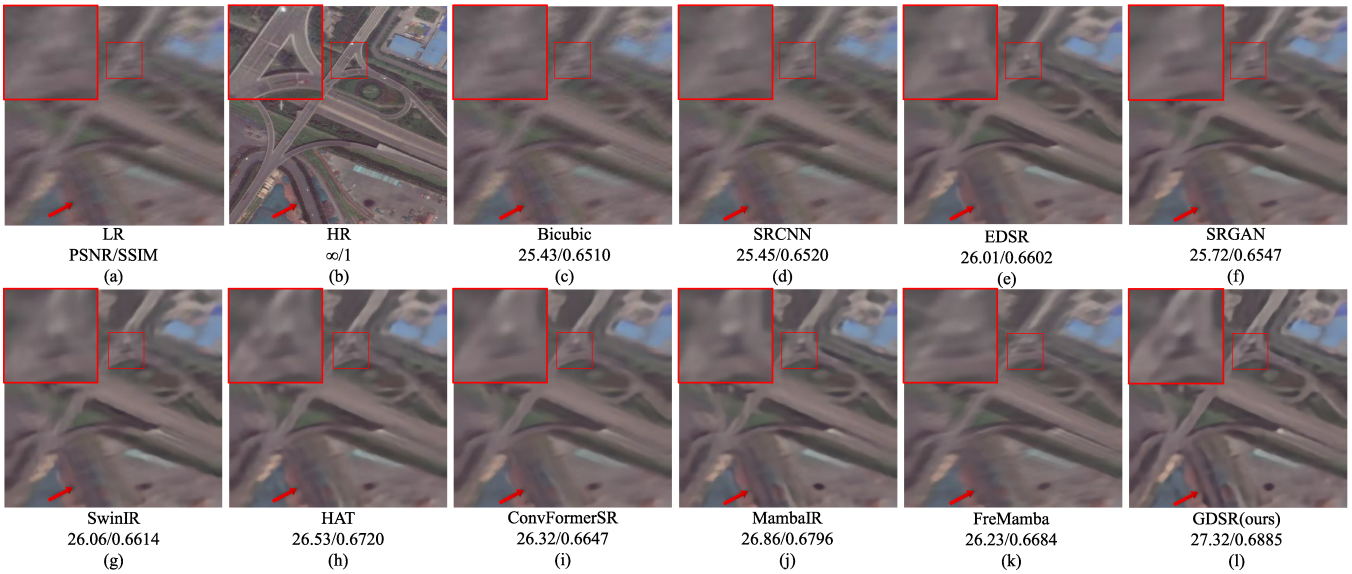

Fig. 11. Visual comparisons of our GDSR with CNN-, GAN-, Transformer-, and Mamba-based methods on the AID_CDM dataset's "viaduct_398" image with scale ×3. Zoomed-in view for better observation.

competing Transformer-based method HAT, the hybrid CNN–Transformer method ConvFormerSR, and the Mamba-based model MambaIR all failed to reconstruct the HR lines on the runway. Only the proposed GDSR accurately restored these details. Furthermore, as illustrated in Fig. 11 for "viaduct_398," under more complex degradation scenarios, GDSR effectively reconstructed the contours, whereas other SR models failed to accurately represent the detailed structure. In Fig. 12, a comparison of the visual results for the RSSRD-QH_CDM dataset showed that the original image, after CDM degradation, appeared heavily blurred and lost its distinctive columnar features. Only the proposed GDSR was able to accurately recover the outline of the original image.

The local attribution map (LAM) comparisons in Fig. 13, based on Figs. 10 and 12, demonstrated that GDSR benefited from the superior long-range modeling capability of RGEG, enabling it to leverage more pixels during the SR process. Additionally, the strong detail extraction capability of RDEG allowed GDSR to utilize pixels correctly for reconstruction. Specifically, for the RSSRD-QH_CDM dataset, when large-scale and global information was present, CNN- and Transformer-based SR networks struggled to extract sufficient information, and while the Mamba-based SR network activated a large number of pixels, it failed to use them correctly for reconstruction. In contrast, thanks to the effective integration of RGEG and RDEG, the GDSR-generated results significantly outperformed other SR models. Moreover, GDSR surpassed MambaIR in terms of a diffusion index (DI) by 25.6193, indicating that by incorporating an RWKV with a larger receptive field into RGEG, GDSR could better capture

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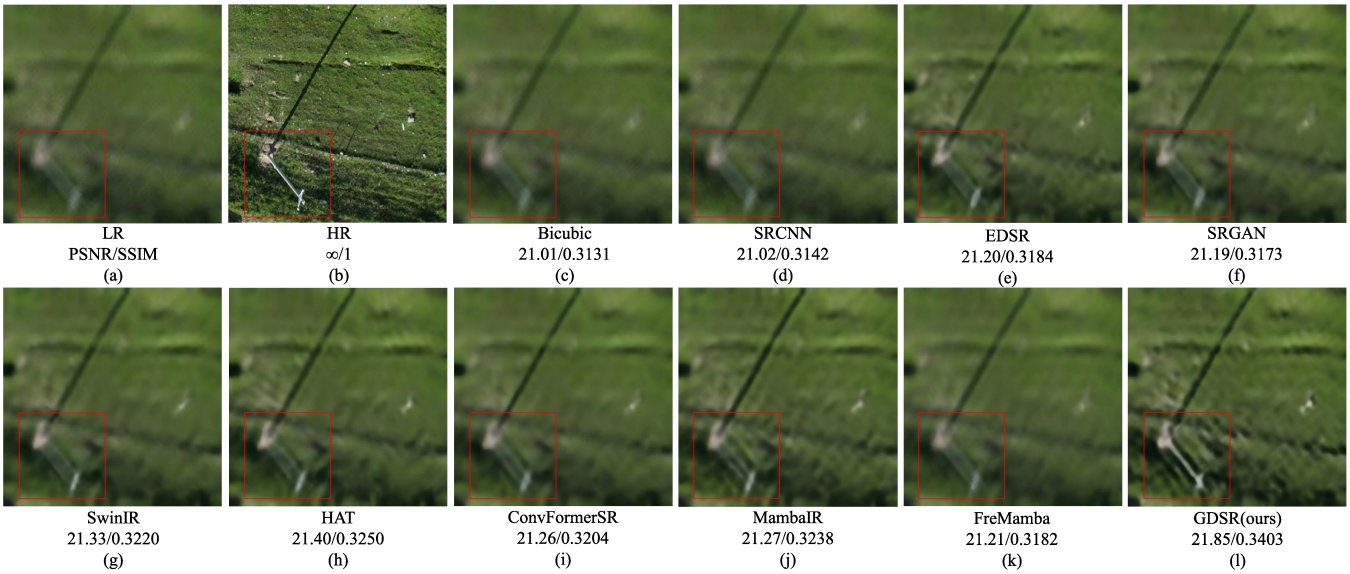

Fig. 12. Visual comparisons of our GDSR with the CNN-, GAN-, Transformer-, and Mamba-based methods on the RSSRD-QH_CDM dataset with scale ×3.

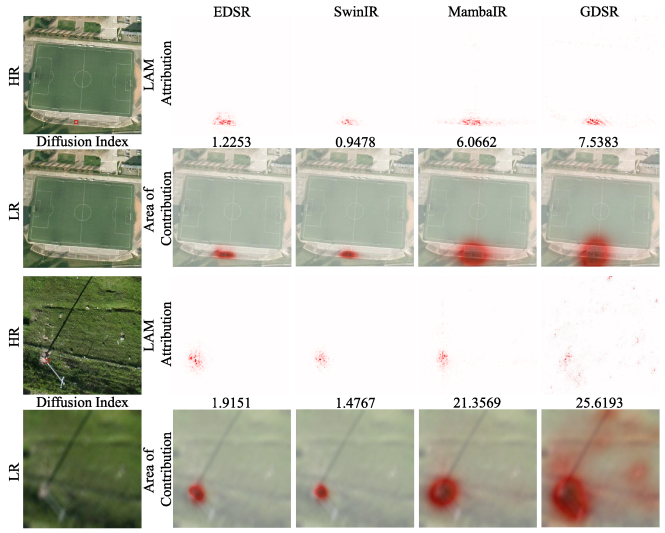

Fig. 13. Visualization of LAM on the AID and RSSRD-QH_CDM datasets.

global information in images with more such features, leading to a higher DI. These visual comparisons further validated the effectiveness of GDSR in capturing and reconstructing details in remote sensing images, highlighting the powerful pixel utilization capability brought by our global-detail dual-branch structure.

5) Real-World Image Testing on the Sanjiangyuan Region: To further evaluate the model's performance in real-world scenarios, we tested it on RSIs from real-world environments. As shown in Fig. 14, the image depicts an alpine meadow grassland type, with surface features such as cracks, scattered rocks, and numerous small "mouse holes." These mouse holes are depressions or openings on the ground, and the distribution of rodents is significant for studying the primary grassland type, the alpine meadow, in Qinghai Province. Their presence can impact the biodiversity, productivity, and ecological function of the alpine meadows. However, due to the limitations in drone flight conditions and resolution, the raw images suffer from motion blur and loss of details. These issues hinder accurate observation of key features such as surface textures and the depth of the mouse holes. By applying SR techniques, our GDSR effectively enhanced the clarity of these fine details, sharpened the edges around the mouse holes, and restored the surface texture. This improvement is crucial for accurate environmental analysis, as the restored landscape provides more precise data for studying biodiversity conservation and the enhancement of ecological functions in the alpine meadow ecosystem of Qinghai Province.

V. Discussion

The proposed method demonstrated consistent SR performance across diverse remote sensing scenarios, validated on the AID (30 land-use categories) [44], UCMerced (21-class geospatial scenes) [45], and custom RSSRD-QH (Qinghai Plateau ecology) datasets. By enhancing texture fidelity in complex environments such as fragmented agricultural landscapes, high-density urban settlements, and ecologically sensitive alpine regions, this approach supports critical applications requiring precise feature extraction. Improved road network reconstruction in severely degraded AID samples aids emergency routing during disasters, while sharpened landscape details in RSSRD-QH enable accurate environmental monitoring. The method's adaptability to both global benchmarks and region-specific challenges positions it as a practical tool for revitalizing historical mid-resolution archives into high-precision analysis-ready data.

Additionally, we tested GDSR's robustness against challenging, remote sensing-specific artifacts. While our initial CDM followed established blind SR methods [49], [50], [51], [52], real-world RSIs often contain severe sensor-induced stripe noise. To validate our method's adaptability, we augmented the CDM with this artifact and retrained the GDSR

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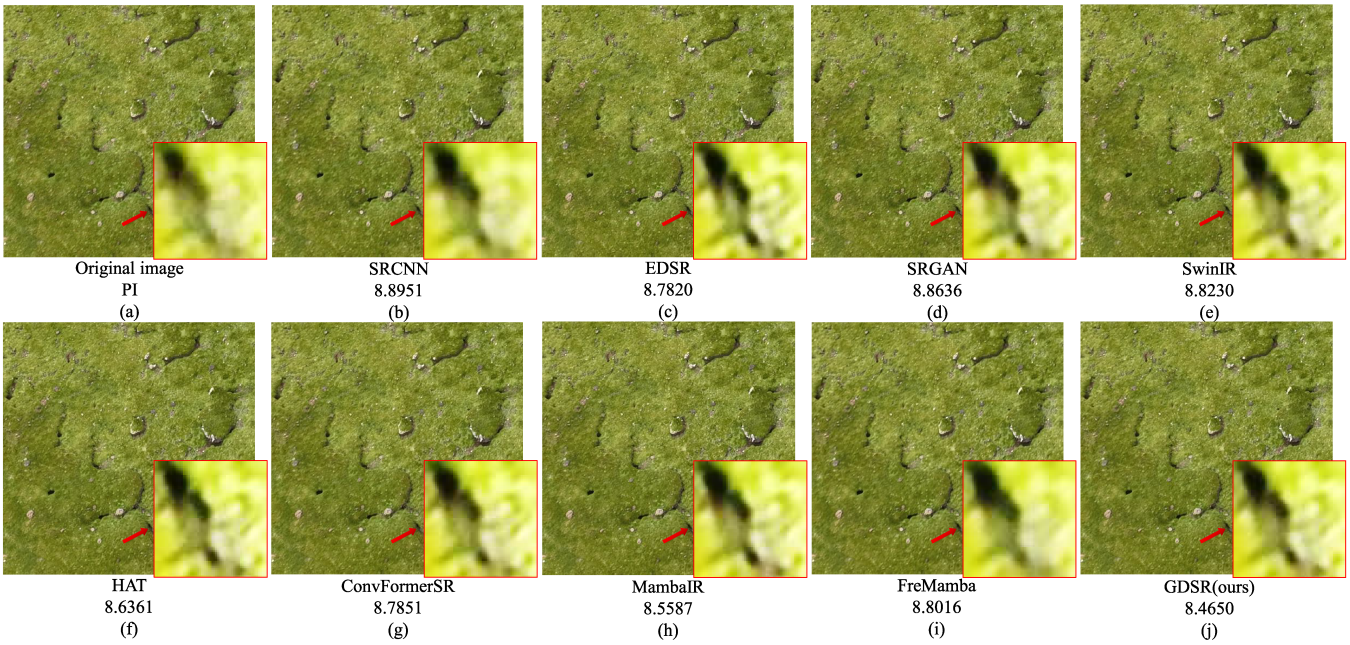

Fig. 14. Visualizations of real-world images for better observation of mouse holes. Zoomed-in view for better observation. We tested the model trained with RSSRD-QH_CDM on real drone images. A lower value indicates better reconstruction quality when using the perception index (PI) metric [54]. Enhanced brightness and contrast for better observation.

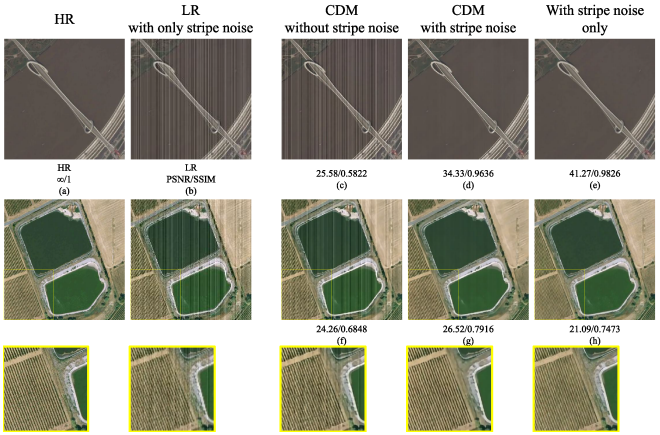

Fig. 15. Visual comparison of GDSR's adaptability to severe remote sensing-specific stripe noise on two samples from the AID dataset. The top and bottom rows show results for "bridge_43" and "pond_410," respectively. (c) and (f) Results from the original GDSR model (trained on CDM without stripe noise) failed to remove the stripes. (d) and (g) Results from the retrained GDSR model (trained on CDM with stripe noise) successfully removed the stripes and reconstructed the scene. (e) and (h) Results from the retrained model on data with only stripe noise, showing its targeted de-striping capability.

model. As demonstrated in Fig. 15, the results confirmed GDSR's strong adaptive capabilities. The retrained model successfully identified and eliminated severe stripe noise in images, leading to quantitative and visual improvements. This positive result paves the way for future work to incorporate other challenging degradations, such as atmospheric scattering, further enhancing the model's practical utility.

VI. Conclusion

In this study, we first introduce the RWKV for RSISR. GDSR effectively models the global dependencies of large-scale RSI while incorporating local details, all with linear complexity and significantly improved inference speed. Specifically, we design an efficient and effective global-detail dual-branch architecture that correctly activates more pixels spatially for SR. To combine global and local representations, we propose GDRM, which adaptively adjusts the features of the dual branches and fuses them. Additionally, we introduce dual-group multiscale wavelet loss to help the model better reconstruct HF details. Extensive quantitative and qualitative experiments on the AID, UCMerced, and RSSRD-QH benchmark datasets using two degradation methods demonstrate that our GDSR outperforms state-of-the-art CNN-, Transformer-, and Mamba-based SR models in RSI-SR tasks.

Nevertheless, our GDSR does exhibit some shortcomings. First, the use of simple stacked residual convolutions for detail extraction introduces a high parameter count, limiting model efficiency. Future work will explore more efficient architectures specifically tailored for RSIs. Second, the current validation is limited to RGB RSIs, lacking evaluation on hyperspectral SR scenarios where balancing spatial enhancement with spectral fidelity is paramount. Extending the method to hyperspectral data and rigorously assessing its spectral accuracy will be a key future direction. Moreover, we consider investigating the model's applicability to multitemporal or cross-sensor RSIs, addressing challenges like temporal consistency and domain adaptation to further enhance its practical utility.

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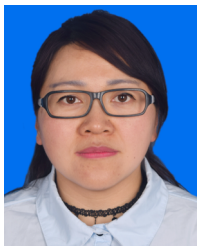

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